

Class Activity — Session 5: Sentiment Analyzer

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Program: BSCS · Section: A · Foundations of Generative AI (CS4106) · Dr. Azhar Dilshad · 30 September 2026

1. Live App URL

Hugging Face Space: <https://huggingface.co/spaces/Jibran101/sentiment-analyzer> — confirmed working (analyzed text live on 30 September 2026).

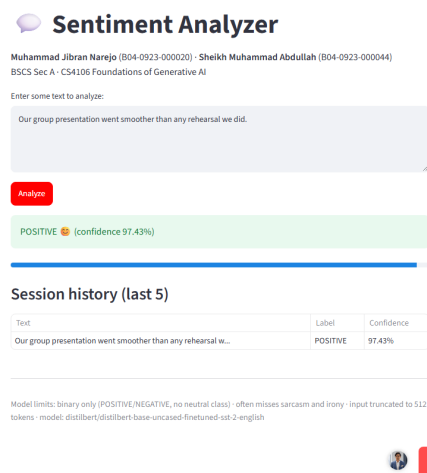
The Space is a free Static Space that embeds the Streamlit Community Cloud app <https://sentiment-analyzer0.streamlit.app/>, because free HF accounts can no longer create Streamlit/Docker Spaces.

2. Commit-History Link

App code: <https://github.com/Jibran-07/sentiment-analyzer/commits/main>

HF Space: <https://huggingface.co/spaces/Jibran101/sentiment-analyzer/commits/main>

3. Screenshots (live app)



Sentiment Analyzer

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BSCS Sec A · CS4106 Foundations of Generative AI

Enter some text to analyze:

Our group presentation went smoother than any rehearsal we did.

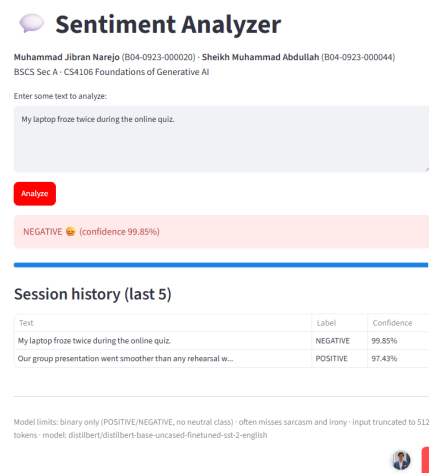
Analyze

POSITIVE 🟡 (confidence 97.43%)

Session history (last 5)

Text	Label	Confidence
Our group presentation went smoother than any rehearsal w...	POSITIVE	97.43%

Model limits: binary only (POSITIVE,NEGATIVE, no neutral class) · often misses sarcasm and irony · input truncated to 512 tokens · model: distilbert/distilbert-base-uncased-finetuned-sst-2-english



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Enter some text to analyze:

My laptop froze twice during the online quiz.

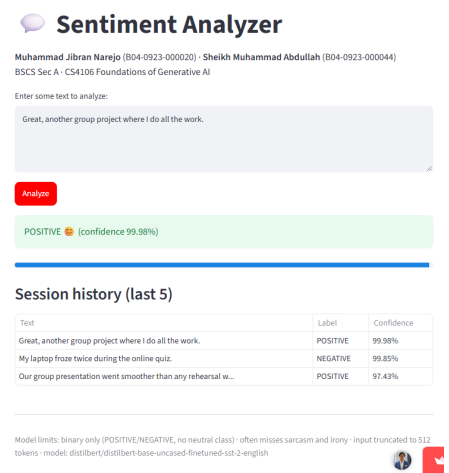
Analyze

NEGATIVE 🟡 (confidence 99.85%)

Session history (last 5)

Text	Label	Confidence
My laptop froze twice during the online quiz.	NEGATIVE	99.85%
Our group presentation went smoother than any rehearsal w...	POSITIVE	97.43%

Model limits: binary only (POSITIVE,NEGATIVE, no neutral class) · often misses sarcasm and irony · input truncated to 512 tokens · model: distilbert/distilbert-base-uncased-finetuned-sst-2-english



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Enter some text to analyze:

Great, another group project where I do all the work.

Analyze

POSITIVE 🟡 (confidence 99.98%)

Session history (last 5)

Text	Label	Confidence
Great, another group project where I do all the work.	POSITIVE	99.98%
My laptop froze twice during the online quiz.	NEGATIVE	99.85%
Our group presentation went smoother than any rehearsal w...	POSITIVE	97.43%

Model limits: binary only (POSITIVE,NEGATIVE, no neutral class) · often misses sarcasm and irony · input truncated to 512 tokens · model: distilbert/distilbert-base-uncased-finetuned-sst-2-english

Positive

Negative

Fools the model

4. Our Personalization

Beyond the starter code we added: a header with both our names and roll numbers; a sidebar *Try an example* button that cycles through campus-themed sample sentences; a session history table (via `st.session_state`) of the last 5 analyses; a low-confidence note below 75%; and a footer listing the model's limits (binary only, sarcasm, 512-token truncation). We also pinned the model, cached it with `@st.cache_resource`, and added truncation and error handling.

5. Final Code

app.py

```
import pandas as pd
import streamlit as st
from transformers import pipeline

MODEL_ID = "distilbert/distilbert-base-uncased-finetuned-sst-2-english"
LOW_CONFIDENCE = 0.75
EXAMPLES = [
    "The cafeteria finally added a vegetarian biryani and it tastes amazing.",
    "The lab printer jammed again ten minutes before the deadline.",
    "Our group presentation went smoother than any rehearsal we did.",
    "The new parking policy is confusing and nobody explained it to us.",
]

st.set_page_config(page_title="Sentiment Analyzer", page_icon="\U0001f4ac")

@st.cache_resource
def load_model():
    return pipeline("sentiment-analysis", model=MODEL_ID)

if "history" not in st.session_state:
    st.session_state.history = []
if "text" not in st.session_state:
    st.session_state.text = ""
if "example_idx" not in st.session_state:
    st.session_state.example_idx = 0

# Sidebar: example filler
with st.sidebar:
    st.header("Try an example")
```

```

st.caption("Fills the text box with a sample sentence.")
if st.button("Load example sentence"):
    st.session_state.text = EXAMPLES[st.session_state.example_idx % len(EXAMPLES)]
    st.session_state.example_idx += 1

# Header
st.title("\U0001f4ac Sentiment Analyzer")
st.markdown(
    """Muhammad Jibrán Narejo** (B04-0923-000020) \xb7 "
    """Sheikh Muhammad Abdullah** (B04-0923-000044) \n"
    """BSCS Sec A \xb7 CS4106 Foundations of Generative AI"
)

text = st.text_area("Enter some text to analyze:", key="text", height=140)

if st.button("Analyze", type="primary"):
    if not text.strip():
        st.warning("Please enter some text first.")
    else:
        try:
            with st.spinner("Analyzing..."):
                result = load_model()(text, truncation=True)[0]
                label = result["label"]
                score = result["score"]
                pct = f"{score * 100:.2f}%"
                if label == "POSITIVE":
                    st.success(f"POSITIVE \U0001f60a (confidence {pct})")
                else:
                    st.error(f"NEGATIVE \U0001f61e (confidence {pct})")
                st.progress(score)
            if score < LOW_CONFIDENCE:
                st.info("Low confidence: the model is unsure about this one, so treat the label with caution.")
                snippet = text if len(text) <= 60 else text[:57] + "..."
                st.session_state.history.insert(0, {"Text": snippet, "Label": label, "Confidence": pct})
                st.session_state.history = st.session_state.history[:5]
            except Exception as e:
                st.error(f"Something went wrong while analyzing: {e}")

if st.session_state.history:
    st.subheader("Session history (last 5)")
    st.table(pd.DataFrame(st.session_state.history))

st.divider()
st.caption(
    "Model limits: binary only (POSITIVE/NEGATIVE, no neutral class) \xb7 "
    "often misses sarcasm and irony \xb7 input truncated to 512 tokens \xb7 "
    f"model: {MODEL_ID}"
)

```

requirements.txt

```

streamlit
transformers
torch

```

index.html (HF Static Space wrapper)

```

<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Sentiment Analyzer</title>
  <style>
    html, body { margin: 0; height: 100%; background: #fff; font-family: system-ui, sans-serif; }
    .bar { padding: 6px 12px; font-size: 13px; color: #555; border-bottom: 1px solid #eee; }
    iframe { border: 0; width: 100%; height: calc(100% - 32px); }
  </style>
</head>
<body>
  <div class="bar">Sentiment Analyzer by Muhammad Jibrán Narejo & Sheikh Muhammad Abdullah (CS4106). If the app is
  asleep, click "Wake up" inside it or <a href="https://sentiment-analyzer0.streamlit.app/" target="_blank">open it
  directly</a>.</div>
  <iframe src="https://sentiment-analyzer0.streamlit.app/?embed=true" title="Sentiment Analyzer" allow="clipboard-
  write"></iframe>
</body>
</html>

```

6. Three Test Cases

Input	Output	Correct?
Our group presentation went smoother than any rehearsal we did.	POSITIVE (97.43%)	Yes
My laptop froze twice during the online quiz.	NEGATIVE (99.85%)	Yes
Great, another group project where I do all the work.	POSITIVE (99.98%)	No, fooled. This is sarcasm: "Great" is a strong positive token and the complaint ("I do all the work") has no explicitly negative word, so the SST-2 model scores it as praise with near-total confidence.

All scores come from real runs (see test_results.txt in the repo). Other finding: "The new library timings made exam week so much easier." was labelled NEGATIVE (98.18%), likely because "exam week" dominates.

7. Reflection

Error we hit: Our Playwright screenshot script worked on localhost but timed out after 180 s on the live Streamlit Cloud URL (`TimeoutError: waiting for get_by_text("Sentiment Analyzer")`). A debug screenshot showed the app was running; Streamlit Cloud serves it inside a nested iframe (`/~/+ /`), so selectors on the outer page found nothing. We fixed it by locating that frame (`page.frames`) and running all selectors against it.

Industry fit: Triageing customer reviews and support tickets, e.g. an e-commerce or telecom team auto-flagging clearly negative feedback so agents respond to it first.

Biggest risk: Confidently wrong labels. The model gave 99.98% to a sarcastic complaint, so decisions made on its score alone (without a neutral class or human review) would silently misread unhappy users.